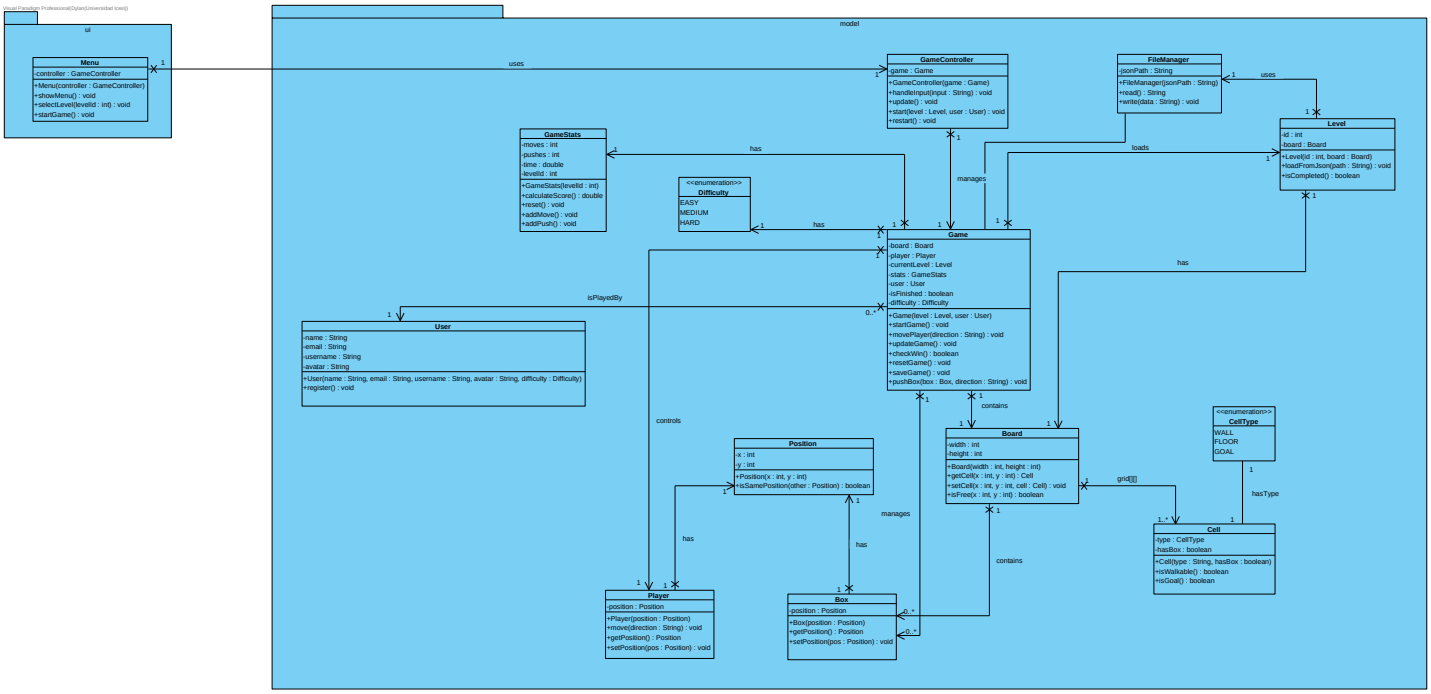
ui
Menu
-controller : GameController
+Menu(controller : GameController)
+showMenu() : void
+selectLevel(levelId : int) : void
+startGame() : void
uses
model
GameController
-game : Game
+GameController(game : Game)
+handleInput(input : String) : void
+update() : void
+start(level : Level, user : User) : void
+restart() : void
FileManager
-jsonPath : String
+FileManager(jsonPath : String)
+read() : String
+write(data : String) : void
uses
Level
-id : int
-board : Board
+Level(id : int, board : Board)
+loadFromJson(path : String) : void
+isCompleted() : boolean
GameStats
-moves : int
-pushes : int
-time : double
-levelId : int
+GameStats(levelId : int)
+calculateScore() : double
+reset() : void
+addMove() : void
+addPush() : void
has
loads
manages
<<enumeration>>
Difficulty
EASY
MEDIUM
HARD
has
Game
-board : Board
-player : Player
-currentLevel : Level
-stats : GameStats
-user : User
-isFinished : boolean
-difficulty : Difficulty
+Game(level : Level, user : User)
+startGame() : void
+movePlayer(direction : String) : void
+updateGame() : void
+checkWin() : boolean
+resetGame() : void
+saveGame() : void
+pushBox(box : Box, direction : String) : void
has
isPlayedBy
User
-name : String
-email : String
-username : String
-avatar : String
+User(name : String, email : String, username : String, avatar : String, difficulty : Difficulty)
+register() : void
contains
controls
Position
-x : int
-y : int
+Position(x : int, y : int)
+isSamePosition(other : Position) : boolean
Board
-width : int
-height : int
+Board(width : int, height : int)
+getCell(x : int, y : int) : Cell
+setCell(x : int, y : int, cell : Cell) : void
+isFree(x : int, y : int) : boolean
grid[][]
<<enumeration>>
CellType
WALL
FLOOR
GOAL
hasType
Cell
-type : CellType
-hasBox : boolean
+Cell(type : String, hasBox : boolean)
+isWalkable() : boolean
+isGoal() : boolean
manages
has
has
contains
Player
-position : Position
+Player(position : Position)
+move(direction : String) : void
+getPosition() : Position
+setPosition(pos : Position) : void
Box
-position : Position
+Box(position : Position)
+getPosition() : Position
+setPosition(pos : Position) : void